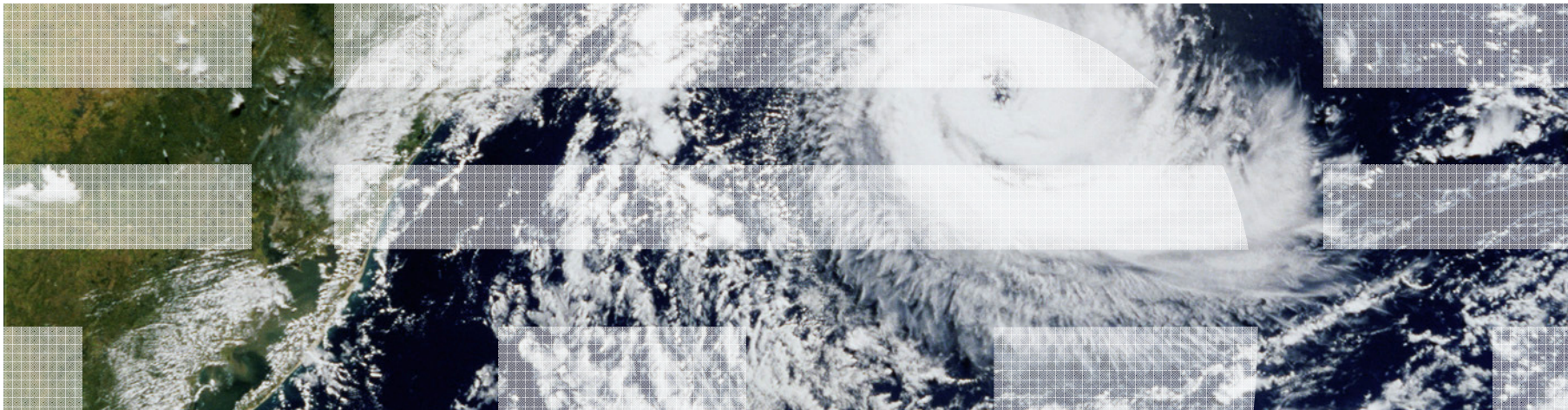


Session 09942: DFSMSrmm: What's new in z/OS V1.13 and z/OS V1.12



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- DFSMShsm
- DFSMSrmm
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- z/OS
- zEnterprise

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Agenda

- z/OS Release 13 Enhancements
- Selected z/OS Release 12 Enhancements
- Appendix

DS8700



**TS1140
Tape Drive**



**TS7680
Deduplication
Gateway**



TS3500 Library



**TS7700
Virtualization
Engine**



Agenda



- z/OS Release 13 Enhancements

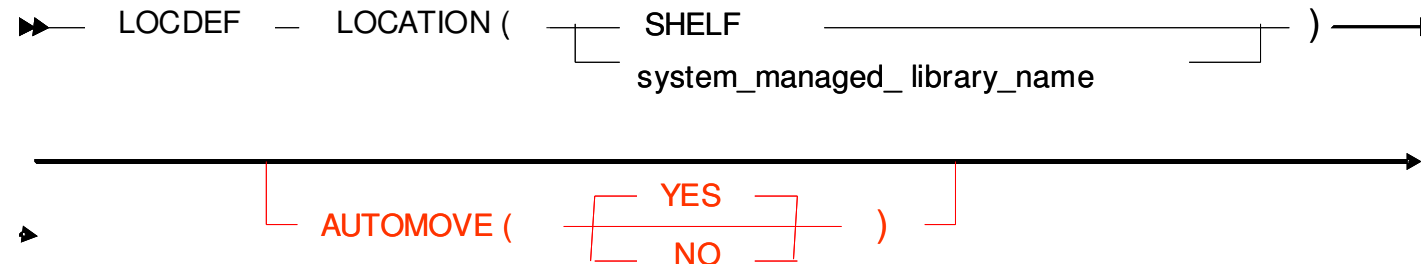


- Selective volume movement
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- Enhanced Tape Copy Support

RMM Selective Volume Movement

- This new capability is designed for libraries that contain virtual volumes or other volumes which either cannot be moved or for which you do not want DFSMSrmm to initiate the movement.
- With new LOCDEF operand AUTOMOVE(YES/NO), you can define locations that are not applicable for automated movement.
- When current location of a volume is defined with LOCDEF...AUTOMOVE(NO), DSTORE processing will not set the destination from the required location.
- During inventory management DSTORE, DFSMSrmm validates the current location name for a volume and determines if automated movement is required.
 - If validation fails no movement is initiated.
 - If a location is not defined via LOCDEF on the inventory management system automated movement is started.
- All volumes can be manually moved by RMM subcommands.

Volume Movement: Parmlib LOCDEF option AUTOMOVE



YES

Volume movement will be attempted by DSTORE processing, if the current location does not match the required location.

NO

No automatic volume movement will attempted.

EDGPC800		DFSMSrmm Location Definitions				Row 1 to 6 of 6	
Command ==>						Scroll ==> PAGE	
Location	Locdef	Management Type	Location Type	Priority	AM	Media Names	
	NO		AUTO	4800	Y		
	NO		MANUAL	4900	Y		
DISTANT	NO		STORE	200	Y		
LOCAL	NO		STORE	300	Y		
REMOTE	NO		STORE	100	Y		
SHELF	NO		SHELF	5000	Y		

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- z/OS Release 13 Enhancements



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More „Last change“ details

- Last change information is added to the
 - TSO list command output
 - ISPF List- , Change- and Delete- panels
 for all resources stored in the RMM CDS
- Reduced need for running EDGAUD audit reports.
- If the most recent change was made by DFSMSrmm processing the ID starts with an asterisk (*).
 - *OAM DFSMSrmm system managed tape support
 - *HKP Inventory management
 - *OCE DFSMSrmm OPEN/CLOSE EOVS support

```

DFSMSrmm Owner Details - BSIN
Command ==> _____

Surname   : Sinram
Forenames : Horst
Department : 3272

Address:
Line 1    : IBM R&D
Line 2    : D-71032 Boeblingen
Line 3    : Germany

Telephone:
Internal  :
External :

Electronic mail:
Email    : sinram@de.ibm.com
Userid   : SINRAM      Node . . . . : IBMDE

Last Change information:
Date . . . . . : 2011/07/05      Time . . : 15:29:17      System : IRD6
User change date : 2011/07/05      Time . . : 15:29:17      User ID : BSIN
  
```

Agenda

- z/OS Release 13 Enhancements

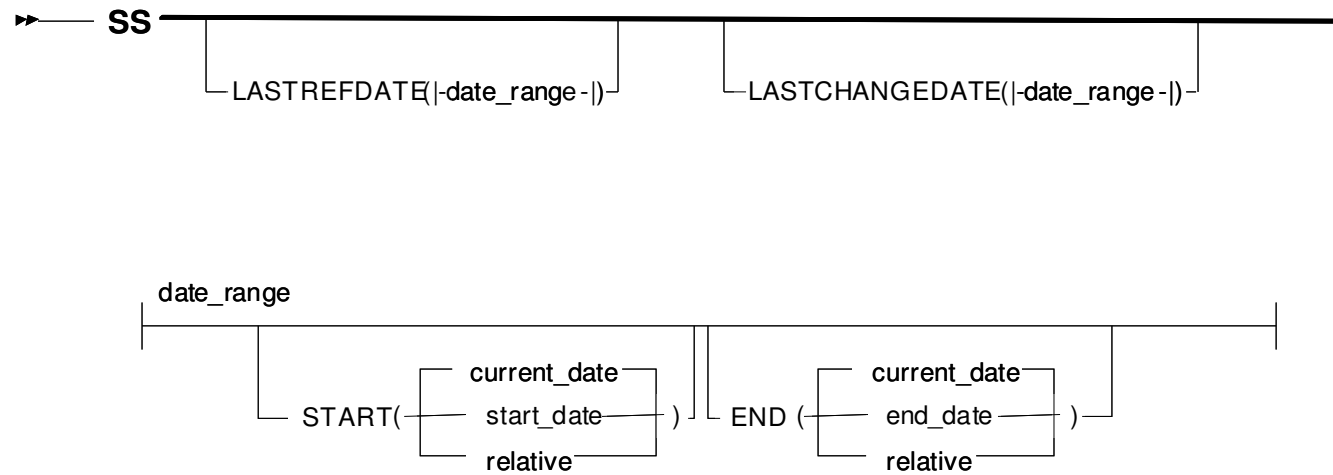


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Last Reference Date for VRS

- After some time the number of Vital Record Specifications in a customer installation may be grown to a number that is hard to comprehend.
 - Especially no longer used VRS' are hard to identify.
- RMM can now
 - display the VRS last reference date in the dialog, and
 - Allow to sorting / search results by last reference date
- You can now cleanup unused VRS' more easily

Last Reference Date for VRS SEARCHVRS



RMM ISPF Panel Updates (1)

```

                                DFSMSrmm Display Data Set VRS
Command ==> _____

Data set mask . : 'BACKUP.DB2ISA.**'                      GDG . : NO
Job name mask . :

Count . . . : 35                      Retention type . . . : DAYS
While cataloged . . . : NO
Delay . . . : 0    Days                Until expired . . . : NO

Location . . . : HOME
Number in location : 35
Priority . . . : 0

Next VRS in chain . :                      Release options:
Chain Using . . . :                      Expiry date ignore . . . : NO
                                      Scratch immediate . . . : NO

Owner . . . : DBUSER
Description . . . :
Last reference : 2011/07/03 16:33:42 ( YYYY/MM/DD HH:MM:SS )
Delete date . . : 1999/12/31 ( YYYY/MM/DD )

Last Change information:
Date . . . : 2011/07/03    Time . : 17:04:00    System : IRD6
User change date : 2011/04/04    Time . : 07:43:27    User ID : DBUSER
  
```

RMM ISPF Panel Updates (2)

DFSMSrmm Search VRSS

Command ==> _____

Optionally specify one of:

Data set mask _____ GDG . . . _____
 (Yes or No)

Job name mask _____

Volume serial _____ Retention type _____

VRs name _____ While cataloged _____ (Yes or blank for all)

Until expired _____ (Yes or blank for all)

Location _____ Release options:

Next VRs in chain _____ Expiry date ignore _____ (Yes or No)

Chain using _____ Scratch immediate _____ (Yes or No)

Owner * _____

Limit 150 Limit search to first n VRSSs. Default is *

Dates Start _____ End _____ Date, date range or relative value

Reference . . . 2011/01/01 . . . _____

Changed . . . -6M . . . _____

Clist NO YES to create a data set, or NO, or blank

DFSMSrmm VRSSs (Page 4 of 4) Row 1 to 30 of 150

Command ==> _____ Scroll ==> PAGE

Enter HELP or PF1 for the list of available line commands.
 Use the LEFT and RIGHT commands to view other data columns.

S	Volume/Data set/Name specification	Count	Store number	Last reference
---	-----	---	---	-----
__	BACKUP.DB2ISA.**	35	35	2011/07/03

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ISPF Navigation Enhancements

- New primary commands CHAINV and CHAIND display multi-volume and multi-file information.
- 16 Point-and-Shoot fields on Volume display, and 5 Point-and-Shoot fields on Data set display allow for more immediate navigation options
 - To control how P&S fields are displayed:
 - Select the Point-and-Shoot... choice from the Colors pull-down, or
 - Issue the ISPF system command PSCOLOR from any ISPF command line
 - Subsequent examples use

Panel Element	Color	Intensity	Highlight	More:
Point-and-Shoot	YELLOW	HIGH	REVERSE	

to highlight P&S fields

ISPF Navigation – New Primary Commands CHAIND/CHAINV

```

DFSMSrmm Data Set Details                                     Multi-File
Command ==> chaind

Data set name . . . : 'D109123.B10501.SEQ003'
Volume serial . . . : B10501      Physical file sequence number . . . : 3
Owner . . . . . : BSIN          Data set sequence number . . . . . : 0
                                           More: +

Job name . . . . . :
Step name . . . . . :
Program name . . . :
DD name . . . . . :
Create date . . . . : 2011/07/05  YYYY/MM/DD
Create time . . . . : 16:45:59
System id . . . . . : IRD6
Expiration date . . : 2011/07/15  YYYY/MM/DD
Set by . . . . . : CMD_DEF

Record format . . . . :
Block size . . . . . : 0
Logical record length : 0
Block count . . . . . : 0
Total block count . . : 0
Data set size (KB) . . : 0
Percent of volume . . : 0
Device number . . . . :

```

The CHAIND
command
shows all data
sets of the multi-
volume set

```

DFSMSrmm Data Sets (Page 1 of 2)                               Row 1 to 11 of 11
Command ==>                                     Scroll ==> PAGE

Enter HELP or PF1 for the list of available line commands
Use the RIGHT command to view other data columns

S  Data set name                               Volume      File
   serial Owner      seq
---
___ D109123.B10501.SEQ001                      B10501 BSIN      1
___ D109123.B10501.SEQ002                      B10501 BSIN      2
___ D109123.B10501.SEQ003                      B10501 BSIN      3
___ D109123.B10501.SEQ004                      B10501 BSIN      4
___ D109123.B10501.SEQ005                      B10501 BSIN      5
___ D109123.B10501.SEQ006                      B10501 BSIN      6
___ D109123.B10501.SEQ007                      B10501 BSIN      7
___ D109123.B10501.SEQ008                      B10501 BSIN      8
___ D109123.B10501.SEQ009                      B10501 BSIN      9
___ D109123.B10501.SEQ010                      B10501 BSIN     10
___ D109123.B10501.SEQ011                      B10501 BSIN     11

```

The CHAINV
command
shows all
volumes of the
multi-volume set

ISPF Navigation Point-and-Shoot Fields

- New point-and-shoot fields are available on the
 - volume and
 - data set display panels

```

                                DFSMSrmm Data Set Details                                Multi-Fi
Command ==> _____

Data set name . . . : 'SAMPLE3.B10001.SEQ002'
Volume serial . . . : B10001      Physical file sequence number . . . : 2
Owner . . . . . : BSIN          Data set sequence number . . . . . : 0
                                                More:

Job name . . . . . :
Step name . . . . . :
Program name . . . :
DD name . . . . . :
Create date . . . . : 2011/07/05  YYYY/MM/DD
Create time . . . . : 16:39:56
System id . . . . . : IRD6

                                Record format . . . . . :
                                Block size . . . . . : 0
                                Logical record length : 0
                                Block count . . . . . : 0
                                Total block count . . : 0
                                Data set size (KB) . . : 0
                                Percent of volume . . : 0
                                Device number . . . . . :

Expiration date . . : 2011/07/15  YYYY/MM/DD
Set by . . . . . : CMD_DEF
Original . . . . . :              YYYY/MM/DD
  
```

Point-and-Shoot in ISPF GUI Client mode

Panel Help

DFSMSrmm Volume Details - A11583 Multi-Vol/Multi-File

Command ==>

Volume	: A11583	VOL1 volser :		Rack number :	R00075
Media name	: 3480	Status . . .	: MASTER	More:	+
Volume type . . .	: PHYSICAL	Stacked count	: 0		
WorldWide ID . . .	:	Worm . . .	: NO		
Retention date . . .	:	Expiration date	: 2010/320		
Set retained . . .	: NO	Set by	: OCE_DEF		
Hold	: NO	Original expiration date . . .	:		
Retention method . .	: VRSEL				

Example of Point-and-Shoot fields on the Display Volume Details panel (GUI mode)

List of Point-and-Shoot Volume Fields

Field

VOL1 volser
Rack number
Set retained
Expiration date
Availability
Owner
Security
Last changed by
Previous volume
Next volume
Volume sequence
Number of data sets
Actions pending
Location
Bin number
Product info

RMM Dialog displays

Volume Details for VOL1 volser
Rack Details
Volume search result list for the volume set
Dialog User Options (date options)
Volume search result list for the volume set
Owner Details
Security Classification Rules for the security level
Owner Details
Volume Details for previous volser
Volume Details for next volser
Volume search result list for the volume set
Data set search result list for all data sets on this volume
Volume Action Status list
Location Definitions list
Bin Details
Product Details

Usage & Invocation – List of Point-and-Shoot Data Set Fields

Field	RMM Dialog displays
Volume serial	Volume Details for volser
Owner	Owner Details
Physical file sequence nbr	Data set search result list for all data sets on this volume
Data set sequence number	Data set search result list for all data sets in the volume set
Create date	Dialog User Options (Date options)

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Show Effective Retention/Expiration Date

- If a resource is retained by VRS, the results list for volumes or data sets might show retained resources with an expiration date that is already passed.
- RMM will now display the **retention date instead of the expiration date** in the
 - volume search results list
 - data set result listif the volume or data set is VRS retained.
- You can now more easily determine from the search results list why a volume is retained, without viewing the volume and data set details.

Show Effective Retention/Expiration Date

- The Search Volume and Search Dataset dialog results list will show the retention date, when a resource is VRS retained.
- The Search Dataset TSO subcommand will return the REXX variables EDG@RTDT and EDG@RTDJ in any case
- The 'Search Dataset' command issued via API will return the RTDJ SFI (X'88C000'), like it was returned by the 'Search Dataset Extended' command before already

Show Effective Retention/Expiration Date Data Set Search Results Panel

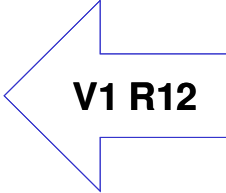
```

Panel  Help  Scroll
-----
EDGPD030      DFSMSrmm Data Sets (Page 2 of 2)      Row 1 to 4 of 4
Command ==>    Scroll ==> PAGE

Enter HELP or PF1 for the list of available line commands
Use the LEFT command to view other data columns

S  Data set name      Create      Expiration V
   -----            date       date       R
   -----            -----
   RMMUSER.D10000      10/04/2010  15/04/2010  Y
   RMMUSER.D20000      10/04/2010  15/04/2010
   RMMUSER.D30000      10/04/2010  15/04/2010  Y
   RMMUSER.D40000      10/04/2010  15/04/2010
***** Bottom of data *****

```



V1 R12

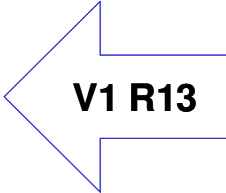
```

Panel  Help  Scroll
-----
      DFSMSrmm Data Sets (Page 2 of 2)      Row 1 to 4 of 4
Command ==>    Scroll ==> PAGE

Enter HELP or PF1 for the list of available line commands
Use the LEFT command to view other data columns

S  Data set name      Create      Expir./      V
   -----            date       Retn. date   R
   -----            -----
   RMMUSER.D10000      10/04/2010  11/04/2010  Y
   RMMUSER.D20000      10/04/2010  15/04/2010
   RMMUSER.D30000      10/04/2010  PERMANENT   Y
   RMMUSER.D40000      10/04/2010  15/04/2010
***** Bottom of data *****

```



V1 R13

Show Effective Retention/Expiration Date Volume Search Results Panel

Panel Help Scroll

DFSMSrmm Volumes (Page 1 of 2) Row 1 to 2 of 2
Command ==> Scroll ==> PAGE

Enter HELP or PF1 for the list of available line commands
Use the RIGHT command to view other data columns

Volume	Assigned	Expiration S	Dest-	Tr- Data
S serial Owner	date	date	ination	ans sets
V10000 RMMUSER	2010/100	2010/105	MASTER SHELF	N 0
V10001 RMMUSER	2010/100	2010/105	VRS SHELF	N 4

***** Bottom of data *****

V1 R12

Panel Help Scroll

DFSMSrmm Volumes (Page 1 of 2) Row 1 to 2 of 2
Command ==> Scroll ==> PAGE

Enter HELP or PF1 for the list of available line commands
Use the RIGHT command to view other data columns

Volume	Assigned	Expir./	S	Dest-	Tr- Data
S serial Owner	date	Retn. date	R Status	Location	ination ans sets
V10000 RMMUSER	2010/100	2010/105	MASTER	SHELF	N 0
V10001 RMMUSER	2010/100	PERMANENT	VRS	SHELF	N 4

***** Bottom of data *****

V1 R13

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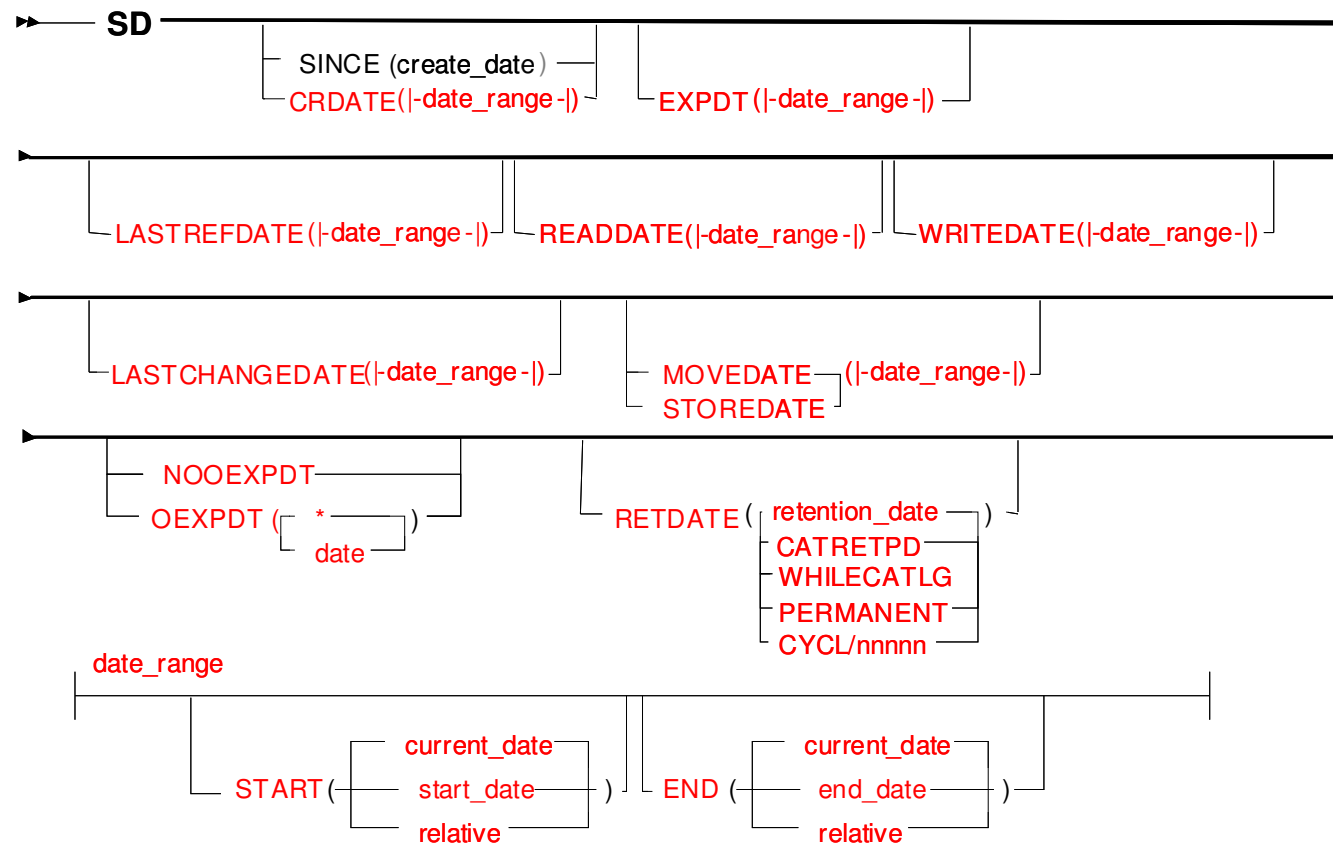
Search Dataset Extensions

- Previously, the SEARCHDATASET subcommand had limited ability for searching on attributes of data sets.
- RMM does now allow to search on more search criteria
 - including many on specific date ranges.
- You can now search more efficiently in a large number of data sets.

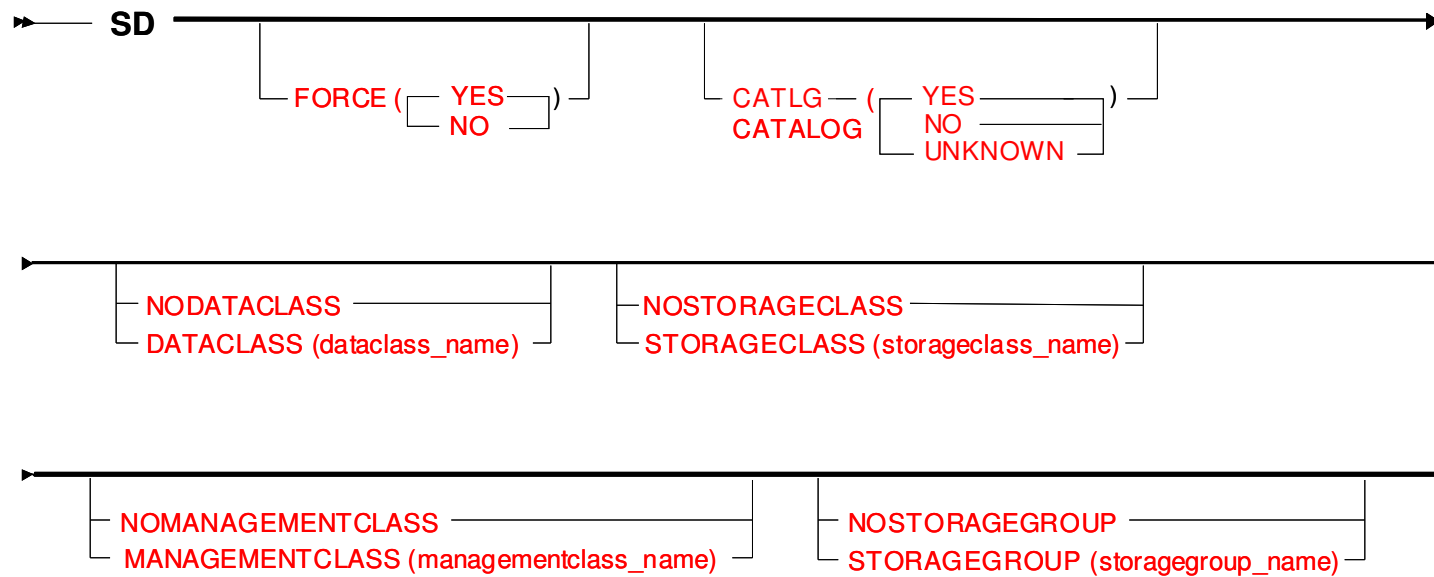
Search Dataset Extensions

SEARCHDATASET Syntax (1/2)

- Options listed in **red** were added:



SEARCHDATASET Syntax (2/2)



Search Dataset Extensions Examples

- List all data sets that were last read or written a month ago or newer:
 - `SD LASTREFDATE(START(-1M)) OWNER(*) LIMIT(*)`

- List all data sets, of which the last CDS change is 1 year ago or longer and that have no original expiration date set:
 - `SD LASTCHANGEDATE(START(1900/001) END(-1Y)) +
NOOEXPDT OWNER(*) LIMIT(*)`

- List all data sets that are retained forever and cataloged:
 - `SD RETDATE(PERMANENT) CATLG(YES) OWNER(*) LIMIT(*)`

- List all data sets, defined with data class “DC000001”, but no storage class:
 - `SD DATACLASS(DC000001) NOSTORAGECLASS OWNER(*) LIMIT(*)`

Usage & Invocation – Dialog

```

DFSMSrmm Data Set Search

Command ==> _____

Enter fully qualified or partial data set name and job name:

Data set name . . . . . _____
Job name . . . . . _____      Specific or generic name

Enter optional parameters to qualify search

Owner . . . . . _____      Owner of volumes (Default is your userid)
Volume serial . . . . . _____      OR Volume serial
List entire set . . . . . _____      YES, For all data sets in the
                                         multi-volume set, otherwise NO
Status . . . . . _____      PRIVATE, SCRATCH, or blank for all
Retained by VRS . . . . . _____      YES, NO, or blank for all
Excluded from VRSEL . . . . . _____      YES, NO, or blank for all

Dates          Start          End          Date, date range or relative value
Create . . . . . _____      . . . . . _____
Expiration . . . . . _____      . . . . . _____
Reference . . . . . _____      . . . . . _____
Read . . . . . _____      . . . . . _____
Write . . . . . _____      . . . . . _____
Changed . . . . . _____      . . . . . _____

Physical file seq . . . . . _____      Relative position on the volume
Limit . . . . . _____ 10 _____      Limit search to first nnnn data sets
Clist . . . . . _____      YES to create a data set, or NO, or blank
Program name . . . . . _____      Specific or generic name
Closed by Abend . . . . . _____      YES, NO, or blank for all
Deleted . . . . . _____      YES, NO, or blank for all
BES key index . . . . . _____      CA BTE tape encryption key index
Original EXPDT . . . . . _____      YES, NO, or a specific date YYYY/MM/DD
Retention . . . . . _____      Data set retained up to YYYY/MM/DD
Cataloged . . . . . _____      YES, NO or UNKNOWN
Force . . . . . _____      Data sets used with FORCE ( YES or NO )
Data class . . . . . _____      Data class name or NO
Storage class . . . . . _____      Storage class name or NO
Management class . . . . . _____      Management class name or NO
Storage group . . . . . _____      Storage group name or NO
  
```

- Also the ISPF panel for SEARCHDATA SET has been enhanced for the new search criteria

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- If DFSMSHsm tapes are expired by the EDGTVEXT HSM exit, extra days for retention can only be defined with an EXTRADAYS VRS.
- With the new parmlib option
TVEXTPURGE(EXPIRE(days))
a number of extra days can be defined globally with no additional VRS definition.

```
>-----+----->
|          .-RELEASE----- .          |
'-TVEXTPURGE (-+-----+-) -'
|          +-NONE-----+
|          .---0--- .          |
'-EXPIRE (-+-days-+-) -'
```

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More information on Expiry Date source

- Just by looking at the expiration date of the volume or data set it may be hard to understand how it was set:
 - Does it stem from OCE or from VOLCAT, during conversion or export, or did RMM set or change it due to parmlib OPTIONS?
- DFSMSrmm now records details of what event caused the EXPDT to be set or changed.
- You can now easily determine what caused the expiration date to be set or changed.

```

Command ==> DFSMSrmm Data Set Details Multi-File

Data set name . . . : 'SAMPLE3.B10001.SEQ002'
Volume serial . . . : B10001      Physical file sequence number . . . : 2
Owner . . . . . : BSIN          Data set sequence number . . . . . : 0
                                   More:      +

Job name . . . . . :
Step name . . . . . :
Program name . . . . . :
DD name . . . . . :
Create date . . . . : 2011/07/05  YYYY/MM/DD  Block count . . . . . : 0
Create time . . . . : 16:39:56                Total block count . . . : 0
System id . . . . . : IRD6                    Data set size (KB) . . . : 0
                                   Percent of volume . . . . . : 0
Expiration date . . : 2011/07/15  YYYY/MM/DD  Device number . . . . . :
Set by . . . . . : CMD_DEF
  
```

More information on Expiry Date source

The **Set by** field displays the event that caused the expiration date to be set or changed:

blank	Not set
CMD	Set by TSO subcommand
CMD_DEF	Default RETPD applied during subcommand processing
CMD_VOLCAT	EXPDT obtained from VOLCAT during subcommand processing
OCE_JFCB	EXPDT obtained from EXPDT/RETPD keywords or from dataclass applied during tape recording
OCE_EXIT	EDG_EXIT100 updated the JFCB EXPDT during tape recording
OCE_DEF	Default RETPD applied during tape recording
OCE_MAX	MAXRETPD was used to reduce the requested EXPDT during tape recording
OCE_VOLCAT	EXPDT obtained from VOLCAT during tape recording
LCS	EXPDT obtained from VOLCAT for system managed tapes when called from OAM installation exits
LCS_DEF	Default RETPD applied for system managed tapes when called from OAM installation exits
TVEXTPURGE	Set as a result of TVEXTPURGE parmlib option
CNVT	Set during conversion by EDGCNVT
EXPORT	Set during export processing

Reporting on Expiry Date source

EDGRDEXT – Data set name record

```

...
RDTOTAL_BLKCNT  DS    0CL20      Total block count across all vol
...
RDESB      DS      CL10      Expdt set by
RDUCDATE DS    CL10      Last "user" change date of
*                               data set record

```

EDGRVEXT – Volume record

```

...
RVHOLD          DS  C          VOLUME HOLD - Y/N
RVESB      DS CL10      Expdt set by
RVUCDATE        DS  CL10      LAST "USER" CHANGE DATE
RVUCTIME        DS  CL6       LAST "USER" CHANGE TIME (HHMMSS)

```

EDGRXEXT – Extended extract record

```

...
XVHOLD          DS  C          VOLUME HOLD - Y/N
XVESB      DS CL10      Expdt set by - of the volume
XDESB      DS CL10      Expdt set by - of the data set
XVUCDATE        DS  CL10      VOLUME LAST "USER" CHANGE DATE
XVUCTIME        DS  CL6       VOLUME LAST "USER" CHANGE TIME

```

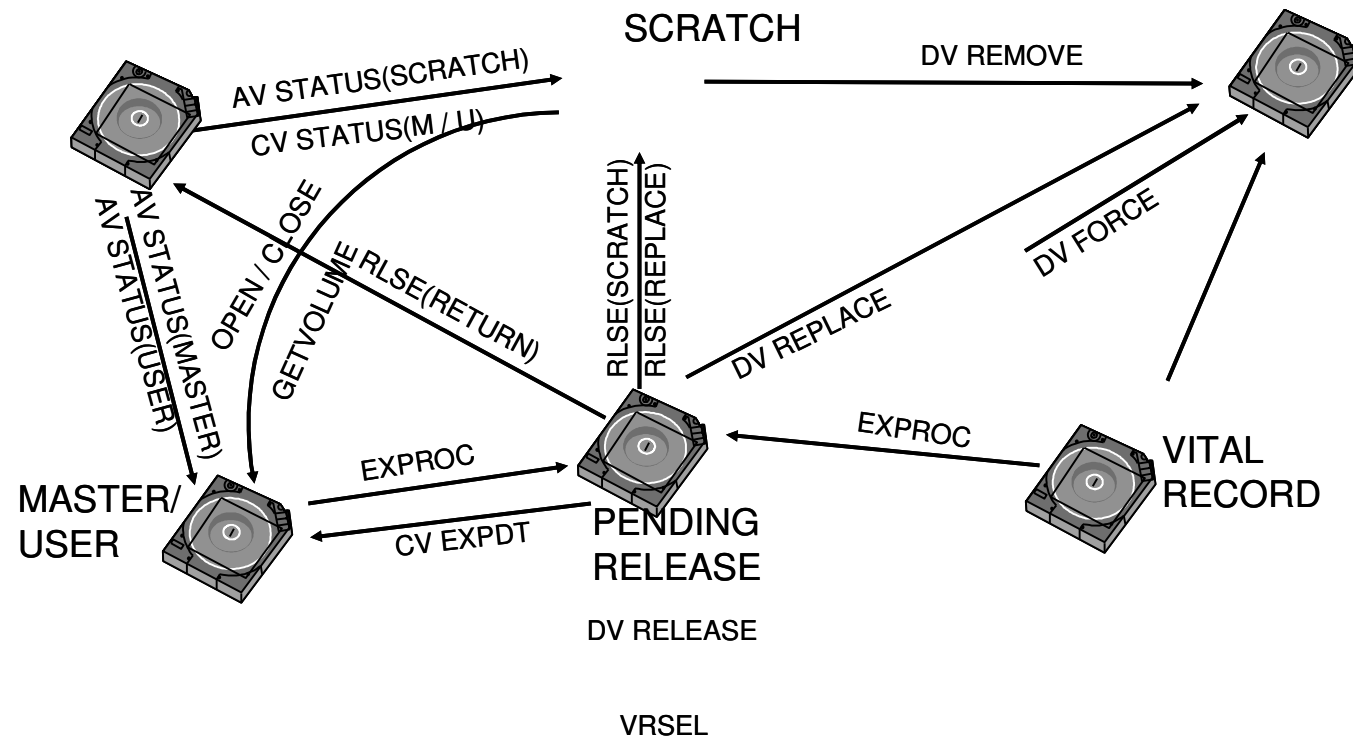
Agenda

- z/OS Release 13 Enhancements
 - Selective volume movement
 - More „Last change“ details
 - Last Reference Date for VRS
 - ISPF Navigation Enhancements
 - Show Effective Retention/Expiration Date
 - Search Dataset Extensions
 - TVEXTPURGE Extra Days
 - More information on Expiry Date source
 - **Exclude data sets from VRSEL**
 - **New RETENTIONMETHOD(EXPDT)**
 - Enhanced Tape Copy Support



Exclude data sets from VRSEL and RETENTIONMETHOD(EXPDT)

- Previously data in the RMM inventory were managed by dynamic VRS policies. With every housekeeping run the retention for a volume or a data set can change.

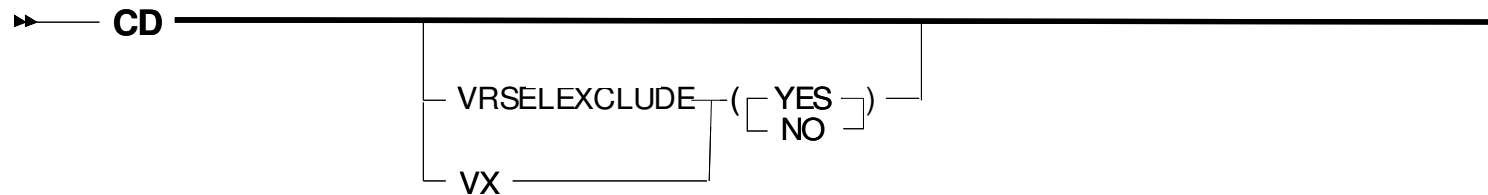


- While VRS' are extremely powerful constructs not all data may require that dynamic management

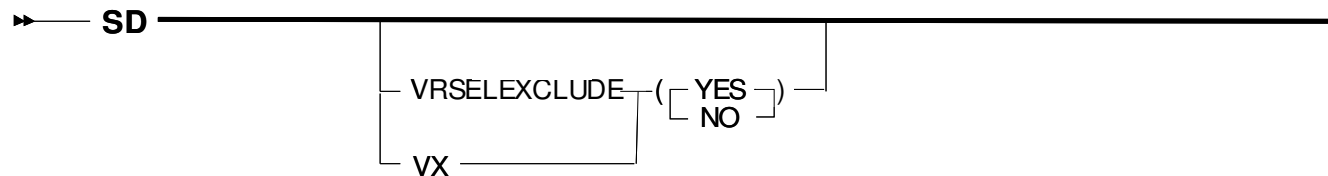
Exclude data sets from VRSEL and RETENTIONMETHOD(EXPDT)

- **It is now possible to**
 - exclude certain data from VRSEL, or to
 - assign a retention method at the time a tape data set is created .
 - This provides you with a choice of whether data is managed by expiration date or by VRS policies.
- With these capabilities you can
 - reduce VRSEL runtime by eliminating certain types of data from VRSEL processing
 - Effectively achieve Controlling Data set Support When a volume or volume set is VRS retained, and you exclude one or more of the data sets from VRSEL, you can have the volume managed just by those data sets not excluded from VRSEL.
 - **Work with simpler retention policies and avoid or reduce batch VRS policy management.**
 - **The retention information for expiration date retained data can be known when a tape data set is created.**
- **Note: z/OS releases lower than V1R13 require the PTF for coexistence APAR **OA32984** to be installed before exploiting these new functions on V1R13.**

Exclude data sets from VRSEL: RMM CHANGEDATASET and SEARCHDATASET Commands



- If VRSELEXCLUDE is changed from NO to YES then:
 - the “VRS retained” flag is reset and
 - the retention date is set to the **current** date
- For multivolume data sets set VX on every data set record



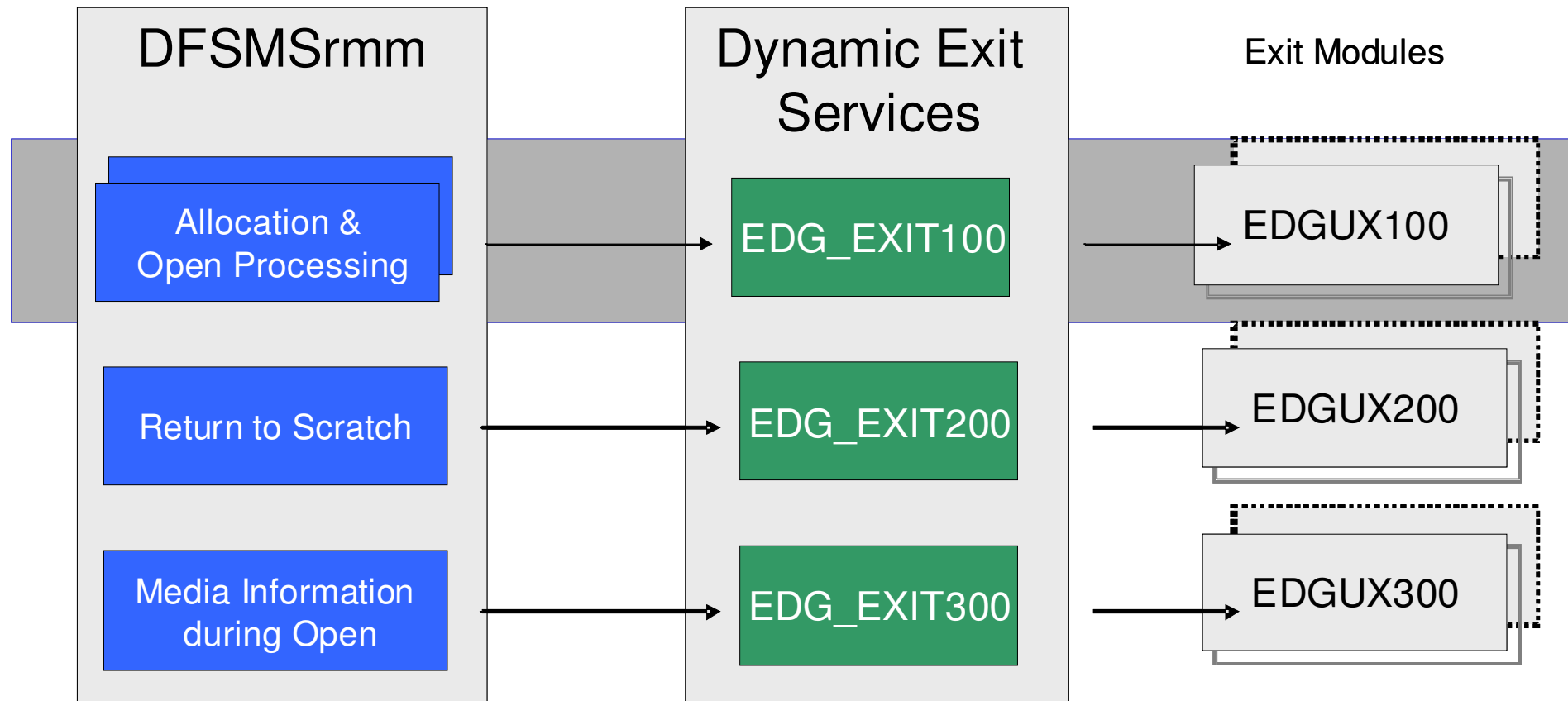
Exclude data sets from VRSEL: ISPF Panel Updates

DFSMSrmm Data Set Details		Multi-File
Command ==> _____		
Data set name	'SAMPLE3.B10001.SEQ002'	
Volume serial	B10001	Physical file sequence number : 2
Owner	BSIN	Data set sequence number : 0
		More: +
Job name		
Step name	Record format :	
Program name	Block size : 0	
DD name	Logical record length : 0	
Create date	2011/07/05 YYYY/MM/DD	Block count : 0
Create time	16:39:56	Total block count : 0
System id	IRD6	Data set size (KB) : 0
		Percent of volume : 0
Expiration date	2011/07/15 YYYY/MM/DD	Device number :
Set by	CMD_DEF	
Original	YY' EDGPD010	DFSMSrmm Data Set Search
Command ==> _____		
Enter fully qualified or partial data set name and job name:		
Data set name ** _____		
Job name _____ Specific or generic name		
Enter optional parameters to qualify search		
More: +		
Owner	BSIN	Owner of volumes (Default is your userid)
Volume serial	_____	OR Volume serial
List entire set	_____	YES, For all data sets in the
		multi-volume set, otherwise NO
Status	_____	PRIVATE, SCRATCH, or blank for all
Retained by VRS	_____	YES, NO, or blank for all
Excluded from VRSEL	yes	YES, NO, or blank for all
Dates	Start _____ End _____	Date, date range or relative value
Create	_____	
Expiration	_____	
Reference	_____	
Read	_____	
Write	_____	
Changed	_____	

Exclude data sets from VRSEL: ISPF Panel Updates Change Data Set

EDGPD310		DFSMSrmm Change Data Set Details	
Command ==> █			
Data set name . . . : 'SAMPLE3.B10001.SEQ001'			
Volume serial . . .	B10001	Physical file sequence number . . .	1
Owner	BSIN	Data set sequence number	0
			More: +
Job name	_____	Record format	_____
Step name	_____	Block size	0
Program name	_____	Logical record length	0
DD name	_____	Block count	0
Create date	2011/07/05	Total block count . . .	0
Create time	16:39:56	Percent of volume . . .	0
System id	IRD6	Device number	_____
Last job name . . . :		Last DD name :	
Last step name . . . :		Last device number . . . :	
Last program name :		VRS management value	
Date last read . . .		Management class . . .	
Date last written		Data class	
VRSEL exclude . . .		Storage class	
Retention date . . .		Storage group	
VRS retained . . .			

Exclude data sets from VRSEL: Installation Exits Overview



Exclude data sets from VRSEL: Exit Support VRSELEXCLUDE

- A new EDG_EXIT100 option is provided to request overriding DFSMSrmm VRSEL processing for specific data sets as they are created or re-written.
- The data set VRSELEXCLUDE attribute is set for all data sets on volumes managed by the EXPDT retention method, and is not affected by this support. If a data set is already retained as a vital record, the vital record attribute is reset and the retention date set to the current date.
- Copy the sample EDGUX100 exit module and use the copy as a base for your exit module.
 - Only perform your processing if the PL100_CAN_VRSELEXCLUDE bit is on.
 - Set PL100_SET_VRSELEXCLUDE bit to B'1' for data sets. If you do **not** request VRSELEXCLUDE the default for the retention method will be used.
If the installation exit sets PL100_SET_VRSELEXCLUDE then any VRS management value set in PL100_VRS is ignored.
 - You do not need to set the PL100_SET_VRSELEXCLUDE bit when you also request to set the retention method to EXPDT: DFSMSrmm always sets the VRSELEXCLUDE attribute for data sets managed by the EXPDT retention method.
- Make any other changes required such as setting the retention method when creating the first file on the tape, or clearing the EXPDT.

Exclude data sets from VRSEL: EDG_EXIT100 Sample VX Table

- The sample EDGUX100 exit module includes an example of setting the VRSELEXCLUDE attribute.
 - The order in which the table entries are listed is important because the exit scans the table until it finds the first entry where the job name, data set name and program name masks match the current request. You can change the priority of matching by changing the order of the table entries.

VXTAB	DS	0F	START OF VRSELEXCLUDE TABLE
	SPACE	1	
	DC	CL8 '*'	JOBNAME
	DC	CL44 'RMMUSER.VX.*'	DATA SET NAME
	DC	CL8 '*'	PROGRAM NAME
	SPACE	1	
	DC	CL8 'VX END'	END OF VX TABLE MARKER

Migration & Coexistence

- There are no migration concerns introduced by this support.
- Standard coexistence recognizes and supports:
 - Data set level VRSELEXCLUDEVRSEL processing on releases below z/OS V1.13 skips these data sets

Agenda

- z/OS Release 13 Enhancements
 - Selective volume movement
 - More „Last change“ details
 - Last Reference Date for VRS
 - ISPF Navigation Enhancements
 - Show Effective Retention/Expiration Date
 - Search Dataset Extensions
 - TVEXTPURGE Extra Days
 - More information on Expiry Date source
 - Exclude data sets from VRSEL
 - **New RETENTIONMETHOD(EXPDT)**
 - Enhanced Tape Copy Support



RETENTIONMETHOD(EXPDT): RMM Parmlib Option (1)

- A new parmlib options RETENTIONMETHOD(EXPDT) allows to set the system-wide default retention method for new tape volume sets created during OCE processing, and for tape volumes added to the DFSMSrmm CDS.

OPTION Command: RETENTIONMETHOD

```
>-----+----->
|                               .-VRSEL-----|
'---RETENTIONMETHOD---(-+-----+)-'
'--RM-----'      '-EXPDT-----'
```

RETENTIONMETHOD(EXPDT): RMM Parmlib Option (2)

■ **VRSEL**

- Specify VRSEL to set the default retention method for new tape volume sets to be VRSEL. This option enables DFSMSrmm inventory management to attempt to match data sets and volumes to VRSEs, and if a match is found, to determine if the data set or volumes is to be retained by VRS.
- The VRSEL retention method is controlled by all the other VRS related options in parmliib including OPTION RETAINBY MOVEBY.

■ **EXPDT**

- Specify EXPDT to set the default retention method for new tape volume sets to be based on EXPDT. Data sets and volumes managed by this retention method are never processed by VRSEL inventory management.
- The EXPDT retention method manages at the volume level: each volume is considered separately for expiration, and each file on a volume can increment the volume expiration date.
- All files of a multi volume data set on a volume set managed by the EXPDT retention method have the same expiration date and time.
- When you specify the EXPDT retention method the DFSMSrmm inventory management EXPROC processing always attempts to return volumes to scratch on the same run as the volume is released (Note: this is as if the SCRATCHIMMEDIATE attribute is set for the volume).

- Default: RETENTIONMETHOD(VRSEL)

- Use parmliib RETENTIONMETHOD(EXPDT) only once you want to switch the default retention method to EXPDT.

RM(EXPDT): ADDVOLUME subcommand

```
>--AV---+-----+-----+-----+-----+-----+-----+-----+-----+----->
|                                     .-VRSEL----- . |
| '---RETENTIONMETHOD--- (-+-----+-----+-----+-----+-----) - '
| '---RM-----' '---EXPDT-----'
```

RETENTIONMETHOD|RM(VRSEL|EXPDT)

- Use this operand to set the retention method for a tape volume set. Specify this operand for the first volume in a multi-volume sequence. All other volumes added to the set assume the same retention method.
- Specify **VRSEL** to set the retention method for a tape volume set to be VRSEL. This option enables DFSMSrmm inventory management to attempt to match data sets and volumes to VRSEs, and if a match is found, to determine if the data set or volumes are to be retained by VRS.
- Specify **EXPDT** to set the retention method for a tape volume set to be based on EXPDT. Data sets and volumes managed by this retention method are never processed by VRSEL inventory management.

Default: RETENTIONMETHOD as specified or defaulted to in parmlib.

RM(EXPDT): CHANGEVOLUME subcommand

```
>--CV---+-----+-----+-----+-----+-----+-----+-----+-----+-----+----->
|                                     .-VRSEL----- . |
| '---RETENTIONMETHOD--- (-+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----) - '
| '---RM-----' '---EXPDT-----'
```

RETENTIONMETHOD|RM(VRSEL|EXPDT)

- Use this operand for any volume to set the retention method for a tape volume set. Specify this operand only for the first volume in a volume set. All other volumes in the set assume the same retention method.
- Specify **VRSEL** to set the retention method for a tape volume set to be VRSEL. This option enables DFSMSrmm inventory management to attempt to match data sets and volumes to VRSEs, and if a match is found, to determine if the data set or volumes are to be retained by VRS.
- Specify **EXPDT** to set the retention method for a tape volume set to be based on EXPDT. Data sets and volumes managed by this retention method are never processed by VRSEL inventory management.
- RETENTIONMETHOD is mutually exclusive with PREVVOL and NOPREVVOL operands.

RM(EXPDT): SEARCHVOLUME subcommand

```

>--SV---+-----+-----+-----+-----+-----+-----+-----+-----+----->
      |                                     .-VRSEL----- .      |
      '---RETENTIONMETHOD---(-+-----+---)-'
      '-RM-----'      '-EXPDT-----'

```

RETENTIONMETHOD|RM (VRSEL|EXPDT)

Use this operand to restrict the returned volumes based on the retention method.

- Specify VRSEL to select volumes with the VRSEL retention method.
- Specify EXPDT to select volumes with the EXPDT retention method.

RM(EXPDT): LISTCONTROL OPTION Output

```
EDGPC200                      DFSMSrmm System Options Display
Command ==>           

                                                                    More:      +

Parmlib suffix . : 02

Operating mode . : PROTECT

Data sets:
  Control . . . : BRMM.MASTER.CDS
  Journal . . . : BRMM.MASTER.JOURNAL
  CDS id . . . : I
  Catalog SYSID : N
  Retention method : VRSEL
  Retention period:
    Default . . . : 10
    Maximum . . . : NOLIMIT
    Catalog . . . : 12      hours

Journal threshold . : 50 %
Journal transaction : NO

SMF:
  System id . . . : IRD6
  Audit . . . : 248
  Security . . . : 249

Report options:
  Lines per page . : 54
  Date format . . . : JULIAN
```

RM(VRSEL) remains the default

RM(EXPDT): EDG_EXIT100 Retention Method Support

- You can use the EDG_EXIT100 installation exit to set the retention method to be used for new tape data. When you create a new tape volume set, or rewrite an existing set from the first file you can override the system default retention method.

```

ENT1ST  DS    0F
* start of RDS entries
    EDGCVRSR DSN=RMMUSER.RMEXPDT.*,          X
        RM=EXPDT,                             X
        RO=NO,                                X
        RETPD=5
    EDGCVRSR DSN=*,                          X
        RM=NONE,                              X
        RO=NO,                                X
        RETPD=5
* start of keyword dates from EDGC5LDR
    EDGCVRSR KEYDATE=98010,                  X
        VRSVAL=D98010
    EDGCVRSR KEYDATE=99000,                  X
        VRSVAL=D99000
    EDGCVRSR KEYDATE=99010,                  X
        VRSVAL=D99010
    EDGCVRSR KEYDATE=99110,                  X
        VRSVAL=D99110
    EDGCVRSR KEYDATE=99201,                  X
        VRSVAL=D99201
ENTLAST  EDGCVRSR DSN='*',RO='NO'

```

RM(EXPDT): Expiry Date Equalization

- RMM maintains a consistent data set expiration date and time for data sets on EXPDT-managed volumes at these times:
 - During O/C/EOV processing.
The expiration time is rounded up to the next whole hour.
 - Minimize I/O to the RMM CDS.
 - This is done for the first data set record of a data set, and only incremented again when data set creation continues onto a new volume and current time exceeds the rounded value.
 - The ASSIGN time remains unchanged
 - When you specify EXPDT/RETPD on ADDDATASET or CHANGEDATASET subcommands.
 - During CHANGEVOLUME PREVVOL
 - When the retention method of the volume set is changed from VRSEL to EXPDT

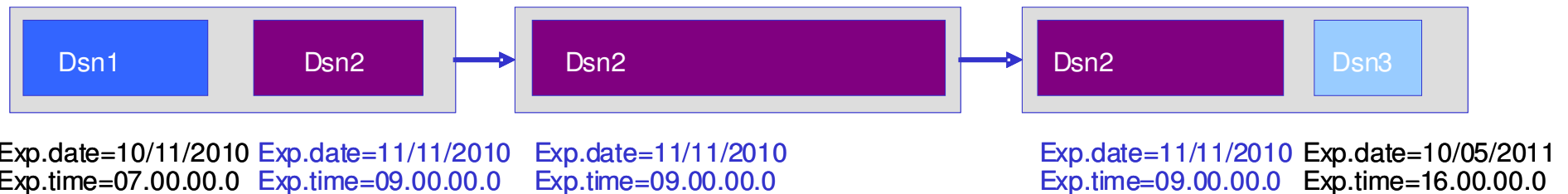
RM(EXPDT): Expiry Date Equalization (cont.)

■ VRSEL retention method

- Data sets on volumes managed by the VRSEL retention method are unchanged.

■ EXPDT retention method

- All files of a multi volume data set on a volume set managed by RM(EXPDT) have the same expiration date and time.



RM(EXPDT): Retention Method Considerations

- You **do not need** to run VRSEL processing unless any volumes are defined with the VRSEL retention method.
 - Only **EXPROC processing is required** to handle expiration of all volumes managed by the EXPDT retention method.
- EXPROC processing provides a summary of volumes by retention method. See the MESSAGE file example below.
- The expiration date of volumes is set during OPEN processing, so for volumes managed by the EXPDT retention method no special considerations exist for open data sets – they are managed based on the volume EXPDT.
- EXPROC processing currently remains to be date-sensitive only (i.e., not time sensitive)
- Volumes containing data sets closed by ABEND processing or which are DELETED are handled as if no special ABEND/DELETED VRS had been defined. i.e. All retention is based only on the volume EXPDT.
- Volumes managed by the EXPDT retention method are included only in the EXPDTDROP limit. VRSRETAIN and VRSDROP limits apply only to volumes managed by VRSEL retention method.

RM(EXPDT): New TSO Subcommand Return/Reason Code

Return Code	Reason Code	Message Number	Issuing Command	Description
12	266	3363	CV, AV	Retention method can only be specified for the first volume in a set

RM(EXPDT): Authorization

Define the resource	To Control the
STGADMIN.EDG.CV.RM ^{minlength}	Updating of retention method. Supporting CV RETENTIONMETHOD RM

When you Define	With Access	Then
STGADMIN.EDG.CV.RM ^{minlength}	Entity not defined	Based on STGADMIN.EDG.MASTER access.
	UPDATE	Allows any volume to be updated

RM(EXPDT): New EDGJRPT/EDGRRPTE Report 18

DFSMSrmm
DATE - 2010/129

Inventory of Data Set Names by Volume Retention Method EXPDT

PAGE - 1 EDGRPT18

TIME - 08:11:53

Data Set Name	Volume Serial	Vol- Seq.	DSN- Seq.	Creating Jobname	Create Date	Create Time	Volume Exp. Date	DSN Exp. Date	V X	EXPDT Set by
RMMUSER.D16002	A16002	1	1		2010/123	081146	2010/133	2010/128	Y	CMD_DEF
RMMUSER.D16003	A16003	2	1		2010/129	081147	2010/135	2010/135	Y	CMD
RMMUSER.D16004.DS1	A16004	3	1	BERNDS	2010/129	081147	2010/134	2010/099	Y	CMD
RMMUSER.D16004.DS2	A16004	3	2	BERNDS	2010/129	081147	2010/134	2010/111	Y	CMD
				End of Report.	4 Entries listed					

DFSMSrmm INTERNAL USE ONLY
EDGRPT18
2010/129

Inventory of Data Set Names by Volume Retention Method VRSEL

PAGE - 2
DATE -

TIME - 08:11:53

Data Set Name	Volume Serial	Vol- Seq.	DSN- Seq.	Creating Jobname	Create Date	Create Time	Volume Ret. Date	DSN Ret. Date	V X	V R
BERNDS.DATASET	A16007	1	1		2010/123	081146	2010/240	2010/240	N	Y
BERNDS.DATASET	A16006	1	1		2010/129	081147	2010/250		Y	Y
BERNDS.DATASET	A16006	1	2		2010/129	081150	2010/250	2010/240	N	Y
RMMUSER.D16001.A	A16001	1	1	TEST	2010/123	081146	PERMANENT	PERMANENT	N	Y
RMMUSER.D16001.B	A16001	1	2	TEST	2010/123	081146	PERMANENT		Y	Y
RMMUSER.D16001.C	A16001	1	3	TEST	2010/123	081146	PERMANENT	PERMANENT	N	Y
RMMUSER.D16005	A16005	1	1		2010/129	081147			N	N

End of Report. 7 Entries listed

RM(EXPDT): New EDGJRPT/EDGRRPTE Report 18

DFSMSrmm
DATE - 2010/129

Inventory of Data Set Names by Volume Retention Method EXPDT

PAGE - 1 EDGRPT18

TIME - 08:11:53

Data Set Name	Volume Serial	Vol- Seq.	DSN- Seq.	Creating Jobname	Create Date	Create Time	Volume Exp. Date	DSN Exp. Date	V X	EXPDT Set by
RMMUSER.D16002	A16002	1	1		2010/123	081146	2010/133	2010/128	Y	CMD_DEF
RMMUSER.D16003	A16003	2	1		2010/129	081147	2010/135	2010/135	Y	CMD
RMMUSER.D16004.DS1	A16004	3	1	BERNDS	2010/129	081147	2010/134	2010/099	Y	CMD
RMMUSER.D16004.DS2	A16004	3	2	BERNDS	2010/129	081147	2010/134	2010/111	Y	CMD

End of Report. 4 Entries listed

DFSMSrmm INTERNAL USE ONLY
EDGRPT18
2010/129

Inventory of Data Set Names by Volume Retention Method VRSEL

PAGE - 2
DATE -

TIME - 08:11:53

Data Set Name	Volume Serial	Vol- Seq.	DSN- Seq.	Creating Jobname	Create Date	Create Time	Volume Ret. Date	DSN Ret. Date	V X	V R
BERNDS.DATASET	A16007	1	1		2010/123	081146	2010/240	2010/240	N	Y
BERNDS.DATASET	A16006	1	1		2010/129	081147	2010/250		Y	Y
BERNDS.DATASET	A16006	1	2		2010/129	081150	2010/250	2010/240	N	Y
RMMUSER.D16001.A	A16001	1	1	TEST	2010/123	081146	PERMANENT	PERMANENT	N	Y
RMMUSER.D16001.B	A16001	1	2	TEST	2010/123	081146	PERMANENT		Y	Y
RMMUSER.D16001.C	A16001	1	3	TEST	2010/123	081146	PERMANENT	PERMANENT	N	Y
RMMUSER.D16005	A16005	1	1		2010/129	081147			N	N

End of Report. 7 Entries listed

RM(EXPDT): EDGJACTP Report

■ VRSRETN Report

- A new data column is added to include the data set VRSEXCLUDE attribute.
- Note that the VRSRETN and VRSRETNS reports are produced only for volumes that are managed by the VRSEL retention method.

■ EXPDROP Report

- A new data column is added to include the retention method

RM(EXPDT): EDGJACTP VRSRTN Report

Newly assigned volumes subject to VRSRETAIN 01/20/09 05:55:21 - 1 -

Status: RETAINED

D A T A S E T				D A T A S E T V R S					V O L U M E				
				V		DROP	REASON			VRS		RETAIN	F
VOLSER	FSEQ	DSNAME	JOBNAME	X	RETAINED	PRIM	2nd.	PRIMARY VRS	JOB MASK	TYPE	VRS	REASON	C
A22251	1	RMMUSER.DSN1		N	Y			RMMUSER.*		D		DATASET	
A22252	1	RMMUSER.DSN20		N	Y			RMMUSER.*		D		DATASET	
A22252	2	RMMUSER.DSN21		N	Y			RMMUSER.*		D		DATASET	
A22253	1	D046059.DSN01		Y	N	W		D046059.*		D		IMPLICIT	
A22253	2	DSN02		Y								IMPLICIT	
A22253	3	D046059.DSN03		N	Y			D046059.*		D		DATASET	
VOL001	1	First.data.set	F1J	N	N	D		ABEND	*	D	VOL001	VOLUME	
VOL001	2	Second.data.set	F1J	Y							VOL001	VOLUME	
VOL002	1	Second.data.set	F1J	Y								SET	
VOL002	2	third.data.set	F1J	Y								SET	

data sets in this status: 10

Newly assigned volumes subject to VRSRETAIN 01/20/09 05:55:21 - 2 -

Status: NOTRETAINED

D A T A S E T				D A T A S E T V R S				V O L U M E					
				V	DROP		REASON	VRS		RETAIN	F		
VOLSER	FSEQ	DSNAME	JOBNAME	X	RETAINED	PRIM	2nd.	PRIMARY VRS	JOB MASK	TYPE	VRS	REASON	C
A22250	1	D046059.WCATALOG	SSTEINHA	N	N	W		D046059.*		D			
A22256	1	DSN6		N									
NO0001	1	ANOTHER.DSET	WOODY1	Y									
NO0001	2	YET.ANOTHER	WOODY2	Y									

data sets in this status: 4

RM(EXPDT): Conversion EDGCNVT SYSIN

■ OPTION RM

- Use the OPTION statement to specify the defaults for DFSMSrmm options which affect volume and data set attributes during conversion.
- Supported options:
 - RM - Specify the **default** retention method. You can override the default retention method at the volume set level by setting the appropriate flags in the 'L record'. See the EDGCLREC mapping: LVRMVRS, LVRMEXP.
- When you specify the RM option you must select one of the following retention_methods:
 - **VRSEL**: Specify VRSEL as the default retention method for all volume sets.
 - **EXPDT**: Specify EXPDT to set the default retention method for volume sets to be based on EXPDT. Data sets and volumes managed by this retention method are never processed by VRSEL inventory management.
 - The EXPDT retention method manages at the volume level: each volume is considered separately for expiration, and each file on a volume can increment the volume expiration date.
- Default: OPTION RM VRSEL

RM(EXPDT): Conversion EDGCNVT Sample JCL

```
// EDGCNVT EXEC PGM=EDGCNVT
// SYSUDUMP DD SYSOUT=*
// SYSPRI NT DD SYSOUT=*
// SYSOUT DD DI SP=(, CATLG), LRECL=80, RECFM=FB,
// SPACE=( 80, ( 1, 1), RLSE), UNI T=SYSDA,
// DSN=D046059. RM EDGCNVT. SYSOUT. DATA
// VRSCMDS DD DI SP=(, CATLG), LRECL=80, RECFM=FB,
// SPACE=( 80, ( 10, 5), RLSE), UNI T=SYSDA,
// DSN=D046059. RM EDGCNVT. VRSCMDS. DATA
// DEXTRCT DD DI SP=SHR, DSN=D046059. RM EDGC5LDR. DEXTOUT. DATA
// DD DI SP=SHR, DSN=D046059. RM EDGC5LDR. DEXTOUTK. DATA
// LI BLI ST DD DI SP=(, CATLG), LRECL=1000, RECFM=VB,
// SPACE=( TRK, ( 1, 1), RLSE), UNI T=SYSDA,
// DSN=D046059. RM EDGCNVT. LI BLI ST. DATA
// OWNLI ST DD DI SP=(, CATLG), LRECL=400, RECFM=VB,
// SPACE=( TRK, ( 1, 1), RLSE), UNI T=SYSDA,
// DSN=D046059. RM EDGCNVT. OWNLI ST. DATA
// BI NLI ST DD DI SP=(, CATLG), LRECL=256, RECFM=VB,
// SPACE=( TRK, ( 1, 1), RLSE), UNI T=SYSDA,
// DSN=D046059. RM EDGCNVT. BI NLI ST. DATA
// VRSLI ST DD DI SP=(, CATLG), LRECL=256, RECFM=VB,
// SPACE=( TRK, ( 1, 1), RLSE), UNI T=SYSDA,
// DSN=D046059. RM EDGCNVT. VRSLI ST. DATA
// SYSI N DD *
OPTI ON RM EXPDT
LOCDEF VAULT1 BI NS
LOCDEF VAULT2 BI NS
IF VMEDIA EQUALS TAPE1600 THEN MEDIA NAME EQUALS REELS
```

Agenda

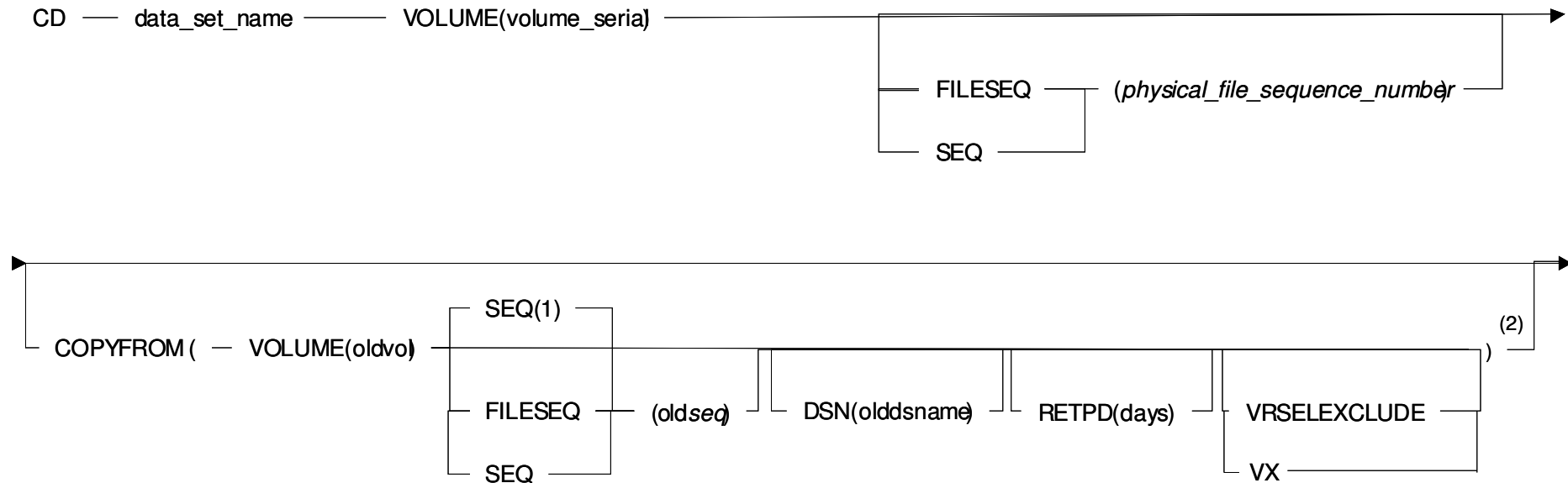
- z/OS Release 13 Enhancements
 - Selective volume movement
 - More „Last change“ details
 - Last Reference Date for VRS
 - ISPF Navigation Enhancements
 - Show Effective Retention/Expiration Date
 - Search Dataset Extensions
 - TVEXTPURGE Extra Days
 - More information on Expiry Date source
 - Exclude data sets from VRSEL
 - **New RETENTIONMETHOD(EXPDT)**
 - Enhanced Tape Copy Support



Enhanced Tape Copy Support

- Tape copy applications could previously use RMM subcommands to update data set and volume meta data after a copy is completed.
 - BUT not all data set and volume attributes could be copied via RMM subcommands.
 - Retention of the source and target data sets is subject to VRS processing and results were not always predictable.
- A new data set subcommand option COPYFROM, does now support copying all applicable attributes and controlling the retention of the source data set.
 - Using installation exit EDG_EXIT100 option tape copy applications can avoid issuing the subcommands needed for copying data set attributes.
- Whether by using subcommands or the user exit copied data sets can now inherit all required attributes
 - Even without the copy application knowing what all the attributes are

Enhanced Tape Copy Support: RMM CHANGEDATASET



- If you specify any other CHANGEDATASET operands
 - the COPYFROM operand is processed first,
 - then the additional operands.
 Therefore additional operands can specify data that overrides the copied attributes.
- Example:
 COPYFROM(RETPD(0) VX)
 copies data set and sets the source data set to pending release

Enhanced Tape Copy Support: Authorization

Define the resource	To Control the
STGADMIN.EDG.CD.COPYFROM. <i>dsname</i> ^{minlength}	Copying of data set attributes from data set <i>dsname</i> to another data set, and to affect the retention of the source data set.

^{minlength}

If you use a generic profile, the minimum non-generic profile name checked for by DFSMSrmm is 'STGADMIN.EDG.CD.'

When you Define	With Access	Then
STGADMIN.EDG.CD.COPYFROM. <i>dsname</i>	Entity not defined	Based on STGADMIN.EDG.MASTER access.
	READ	You are permitted to copy attributes and update retention for identically named data sets.
	UPDATE	You are permitted to copy attributes and update retention for any two data set records.

Enhanced Tape Copy Support: Copying Data Set Attributes

- After copying the data set attributes, all data set records of the target data set make the data set appear to be the original.
- DFSMSrmm copies **all** attributes that are **not related to the physical aspects** of the data set, volume and tape drive.
- Attributes related to retention are subject to update by the next run of inventory management. The intention is that the copied data set will be retained in the same way as the source data set.
- After the attributes are copied there will be no trace of the copy application or the batch job used to perform the copy because all target data set attributes reflect the creation and use of the source data set.
 - The “last change” information of the target data set is updated during command processing to reflect that the command was processed.

Enhanced Tape Copy Support: Data Set Attributes Not Copied

Command operand	Extract file field	REXX Variable/SFI
No cmd operand		
dsname	RDDSDNAME	EDG@DSN
VOLUME	RDVOLSER	EDG@VOL
SEQ FILESEQ	RDDSNSEQ	EDG@FILE
LABELNUMBER	RDLABNO	EDG@DSEQ
TOTALBLKCOUNT	RDTOTAL_BLKCNT	EDG@BLKT
PERCENT	RDPERCENT	EDG@DPCT
DEVNUM	RDUNITAD	EDG@DEV
LRECL	RDLRECL	EDG@LRCL
RECFM	RDRECFM	EDG@RCFM
BLKSIZE	RDBLKSZ	EDG@BLKS
BLKCOUNT	RDBLKCNT	EDG@BLKC
owner	RDOWNDSN	EDG@OWN
data set size	RDDSSIZE,RDSIZE	EDG@DSS6
catalog status	RDCAT	EDG@CTLG
STORAGECLASS	RDSCNAME	EDG@SC
storage group	RDSGNAME	EDG@SG
DATACLASS	RDDCNAME	EDG@DC
start block ID	n/a	n/a
end block ID	n/a	n/a
last device number	RDLDEVN	EDG@LDEV
BESKEY	RDBESKEY	EDG@BESK
VRSELEXCLUDE	RDVEX	EDG@VEX Note: This attribute is not copied unless both the source and target volumes are managed by RM(VRSEL)
ABEND	RDABEND	EDG@ABND Note: This attribute is not copied unless it is set. The source setting is merged with the target setting.

Enhanced Tape Copy Support: Installation Exit Support

- A new EDG_EXIT100 option is provided to notify RMM that the data set being created **is being copied from another**.
- During OPEN processing the exit can identify the source data set from which RMM will obtain all existing data set attributes which will be used for the new data set.
 - RMM EOV processing ensures that the attributes are copied to all new data set records when the output data set becomes a multi-volume data set.

```
PL100_CAN_COPYFROM      EQU X'08'  
PL100_SET_COPYFROM      EQU X'04'  
PL100_COPYFROM_DSN      DS CL44  
PL100_COPYFROM_VOLSER   DS CL6  
PL100_COPYFROM_DSEQ     XL4  
PL100_COPYFROM_OWNER    DS CL8    ...
```

Enhanced Tape Copy Support: [Installation Exit Exploitation](#)

■ **Dynamic Exit Available since z/OS V1R11**

- Add exit module dynamically prior to the first copy
 - Pass source data set key via exit - Dsname, file sequence, volser
 - Select required retention method and VRSELEXCLUDE for target via exit
 - During OPEN &CLOSE the attributes are copied from source to target Data Set record
 - Physical attributes are set based on volume and drive used
- Delete exit module at end of processing / all copies completed
- Your copy application must communicate with your exit module
- When processing successful use the RMM API to:
 - Issue RMM commands for source data sets and volumes, for example:
RMM DV volser RELEASE
or
RMM CV volser RETENTIONMETHOD(EXPDT) RETPD(4)
or
RMM CD sourcedsn VOLUME(sv) SEQ(ss) VRSELEXCLUDE(YES)

Agenda

- z/OS Release 13 Enhancements
- Selected z/OS Release 12 Enhancements
- Appendix

DS8700



**TS1140
Tape Drive**



**TS7680
Deduplication
Gateway**



TS3500 Library



**TS7700
Virtualization
Engine**



Support for TS1140

- DFSMSrmm Support provided via
 - APAR OA35804 (z/OS V1.10)
 - APAR OA33958 (z/OS V1.11 – V1.12)
 - z/OS V1.13
- DFSMSrmm support in z/OS V1.10 and V1.11 is a subset of the SUPPORT available with V1.12 and above

z/OS V1.10 and above provides	In addition, z/OS V.12 and above provides
▪New Media Types ▪New Recording Formats ▪Changed RMM commands: –ADDVOLUME –CHANGEVOLUME –SEARCHVOLUME	▪New maximum values for –BLKCOUNT –TOTALBLKCOUNT ▪New volume and data set attributes ▪LISTVOLUME output; LISTDATASET output ▪RMM ISPF Dialog ▪REXX and API SFIs ▪Reporting

New Media Types and Recording Modes

Media Type	Alias	Full Media Name
MEDIA11	EATC	IBM Enterprise Advanced Tape Cartridge 3592.
MEDIA12	EAWTC	IBM Enterprise Advanced WORM Tape Cartridge 3592.
MEDIA13	EAETC	IBM Enterprise Advanced Economy Tape Cartridge 3592.

Recording Format	Description	Valid with media types
EFMT4	Data is written to the volume in EFMT4 format.	Existing: EXTC, EXWTC, New: EATC, EAWTC, EAETC
EEFMT4	Data is written to the volume in <u>encrypted</u> EFMT4 format.	Existing: EXTC, EXWTC, New: EATC, EAWTC, EAETC

- New types and recording formats can be used with ADDVOLUME, CHANGEVOLUME, SEARCHVOLUME commands.

Larger counts and new attributes

- Larger block counts
 - BLKCOUNT can be now up to 4294967295 (10 characters).
 - ADDDATASET and CHANGEDATASET
 - TOTALBLKCOUNT can be now up to 18446744073709551615 (20 characters)
 - CHANGEDATASET

- New attributes
 - VOLUME
 - Physical space Used
 - Compression
 - DATASET
 - Physical Size
 - Compression

```

EDGPT110          DFSMSrmm volume Details - JC0176
Command ==>

Volume . . . . . : JC0176      VOL1 volser :      Rack number :
Media name . . . . : 3480              Status . . : USER
                                           More:      - +

Volume use count . : 1           Volume usage (KB) . . . . : 500000
Compression . . . : 1.05         Physical space used (KB) . : 476844
Capacity (MB) . . : 3814697      Percent full . . . . . : 0
Create date . . . : 2011/187     Create time . . . . . : 06:36:54
                                   System ID . . . . . : W98MVS2
Date last read . . : 2011/187    Date last written . . . : 2011/187
Drive last used . . : 0BB3       Write Mount count . . . : 1
  
```

$$\text{Compression} = \frac{\text{ApplicationBytesWritten}}{\text{DeviceBytesWritten}}$$

-Due to granularity of data the compression value for smaller data sets may not be meaningful

-0.00 displayed when no ration can be computed

Summary of z/OS DFSMSrmm V1R12

Ease of Use

- Retention Limit Reporting (Roll-back to R10 via APAR OA30881)
- Ignore for duplicate volumes
- Automation for WTORs in Production and Parallel Running
- Expiration Override for Volumes
- ISPF Dialog CLIST option to avoid search results list

Optimization

- Copy Export Sample Reports from Export Status and BVIR

Performance and Scalability

- All RMM Data Sets EAS Eligible, 'XTIOT' Support
- IPV6 Support
- Contribution to z/OS Target of 5% improvement / Release

Availability and RAS

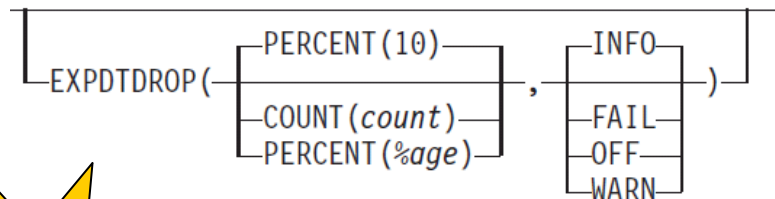
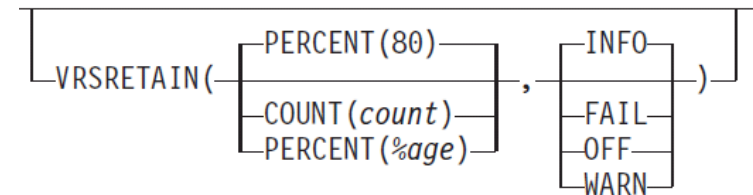
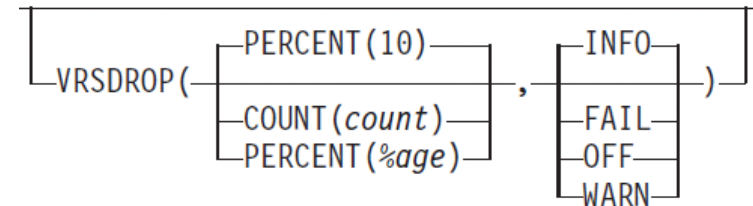
- Status Available via Subcommand and API
- STOP/CANCEL Recovery Improvements
- PDA Trace Enhanced to trace outside subsystem address space

Safety net: EXPDTDROP / VRSDROP / VRSRETAIN

- **VRSDROP** to specifies how many existing VRS-retained volumes may be dropped from vital records retention and the action to be taken by DFSMSrmm.
- **VRSRETAIN** specifies how many newly assigned volumes are to be retained by vital records retention.
 - A newly assigned volume is one that has a volume assignment time that is higher than the run time of the previous VRSEL processing and that is not VRS-retained.
- **EXPDTDROP** specifies how many existing expiration date retained volumes may be dropped from retention.

An EXPDT-retained volume is one that is not VRS-retained and is not newly assigned

 - EXPROC: additional processing may be required



OA33526
required

Sample VRSRETAIN Report

Newly assigned volumes subject to VRSRETAIN										01/20/09	05:55:21	- 1 -	
Status: RETAINED													
D A T A S E T				D A T A S E T		V R S		V O L U M E					
VOLSER	FSEQ	DSNAME	JOBNAME	RETAINED	DROP PRIM	REASON 2nd.	PRIMARY VRS	JOB MASK	VRS TYPE	VRS	RETAIN REASON	FILE COUNT	IN SET
VOL1	1	RMMUSER.DSN11		Y			RMMUSER.*		D		DATASET	3	N
VOL1	2	RMMUSER.DSN12		N	W		RMMUSER.*		D		IMPLICIT	3	N
VOL1	3	RMMUSER.DSN13		Y			RMMUSER.*		D		DATASET	3	N
VOL2	1	D046059.DSN21		N	W		D046059.*		D	VOL2	VOLUME	2	N
VOL2	2	D010155.DSN22								VOL2	VOLUME	2	N
VOL6	1	D046059.DSN61		N	D		D046059.*		D		SET	1	Y
VOL7	1	D077077.DSN71									IMPLICIT	2	Y
VOL7	2	D077077.DSN72		Y			D077077.DSN72		D		DATASET	2	Y
data sets in this status:				8									
Newly assigned volumes subject to VRSRETAIN										01/20/09	05:55:21	- 2 -	
Status: NOTRETAINED													
D A T A S E T				D A T A S E T		V R S		V O L U M E					
VOLSER	FSEQ	DSNAME	JOBNAME	RETAINED	DROP PRIM	REASON 2nd.	PRIMARY VRS	JOB MASK	VRS TYPE	VRS	RETAIN REASON	FILE COUNT	IN SET
VOL3	1	RMMUSER.DSN31	STEINHA	N	W		RMMUSER.*		D			1	Y
VOL4	1	D010155.DSN41										2	Y
VOL4	2	RMMUSER.DSN42		N	W		RMMUSER.*		D			2	Y
VOL5	1	D010155.DSN51										3	N
VOL5	1	D010155.DSN52										3	N
VOL5	1	D010155.DSN53										3	N
data sets in this status:				6									

Summary of newly assigned volumes for VRSRETAIN										01/20/09	05:55:21	- 1 -
Status		VOLUME COUNT										
RETAINED		4										
NOTRETAINED		3										

OPENRULE IGNORE (V1.12)

- Ignore processing for specific volser request (read and write)
 - The external/vision volser is used to identify the mounted volume

Library	OPENRULE for		Result in	
	requested volser	VOL1 volser	V1R11	V1R12
Non-system managed (SHELF)	IGNORE	ACCEPT	requested volser ignored (at file validation)	requested volser ignored (at mount verification)
	ACCEPT	IGNORE	VOL1 volser ignored, LBL ERR VOL1	accepted
System managed (ATL)	IGNORE	ACCEPT	requested volser ignored (at file validation)	requested volser ignored (at mount verification)
	ACCEPT	IGNORE	VOL1 volser ignored, 613-1C requested volser	accepted

Expiration Override

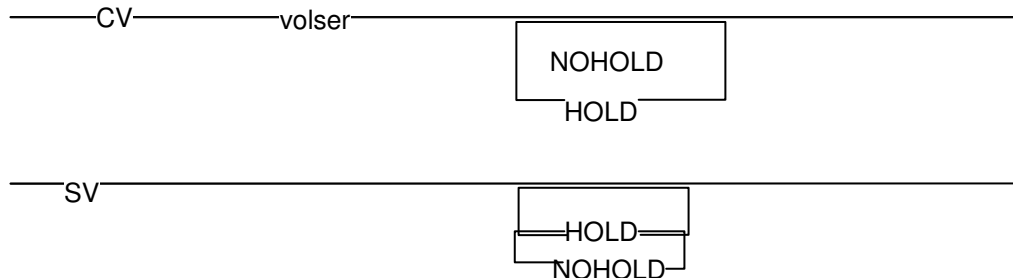
- **Problem**

- Difficult to determine how data sets and volumes are retained
- Setting EXPDT(99365) might not force retention
- Changing policies or EXPDT causes loss of actual retention policy

- **Solution**

- New Subcommand to prevent expiration
- Existing policy and retention is unchanged

- **Subcommand Changes:**



- **Setting using the dialog**

- New 'HY' and 'HN' line commands

- **When HOLD attribute is set:**

- Unable to RELEASE the volume
- EXPROC prevents expiration

Dialog CLIST Option

```
EDGP@CLS                      DFSMSrmm CLIST Processing
Command ==>

Enter optional prefix and suffix values
Prefix . . . . . 'RMM LV '
Returned text depending on resource being searched
Suffix . . . . . ' ALL'

Enter optional fully qualified or partial data set information for CLIST
Data set name . . . .
Expected data set size          records
Extend existing CLIST          YES, NO or blank

View search results   NO          YES, NO or blank
Press ENTER to CONTINUE, or END to RETURN.
```



If you choose **View search results: NO** (which is the default),
the search result list is not displayed

Copy Export processing (v1.12)

- Use EDGJCEXP sample job to generate reports about copy exported data combining information from
 - TS7700 library
 - BVIR or Export status file
 - DFSMSrmm extended extract records
- Reports are provided sorted
 1. by data set name
 2. by logical volume serial number
 3. by stacked volume serial number

TS7700 Virtualization Engine



Copy Export Reporting

- EDGJCEXP Report – sorted by dataset

Copy Exported Data Sets

- 1 -

12/08/2009

03:30:21

based on Bulk Volume Information Retrieval data

DATA SET INFORMATION

DATA SET NAME	CREATE DATE	CREATE TIME	REC FM	BLK SIZE	RETENTION DATE	EXPIRATION DATE	PHYSICAL FILE	V SEQ	R
BERNDS.EXPIRED.HYD868	2009/338	082750	F	80	2009/353	2009/341	1	Y	
BERNDS.EXPIRED.HYD880	2009/337	150732	F	80	2009/352	2009/340	1	Y	
BERNDS.MULTI.VOLUME.DS1	2009/338	082524	FB	80	2009/353	2009/341	1	Y	
BERNDS.MULTI.VOLUME.DS1	2009/338	082524	FB	80	2009/353	2009/341	1	Y	

LOGICAL VOLUME INFO				STACKED VOLUME INFO				COPY EXPORT INFO		
VOLSER	VOLSEQ	REQUIRED LOCATION	EXPIRATION DATE	VOLSER	CURRENT LOCATION	DESTI NATION	IN TRAN	RETENTION DATE	V EXPORT DATE	EXPORT TIME
HYD868	1	MAZ2	2009/341	A02039	ATL3484F	MAZ1	Y	2020/001	Y 2009/338	083938
HYD880	1	MAZ2	2009/341	A02039	ATL3484F	MAZ1	Y	2020/001	Y 2009/338	083938
HYD862	1	MAZ2	2009/341	A02039	ATL3484F	MAZ1	Y	2020/001	Y 2009/338	083938
HYD861	1	MAZ2	2009/341	A02039	ATL3484F	MAZ1	Y	2020/001	Y 2009/338	083938

■ EDGJCEXP Report – sorted logical volume

Copy Exported Data Sets By Logical Volume

- 1 -

12/08/2009

03:30:22

based on Bulk volume Information Retrieval data

Logical volume Info: HYD861 1 MAZ2 2009/341

DATA SET INFORMATION

DATA SET NAME	CREATE DATE	CREATE TIME	REC FM	BLK SIZE	RETENTION DATE	EXPIRATION DATE	PHYSICAL FILE	SEQ	V R
BERNDS.MULTI.VOLUME.DS1	2009/338	082524	FB	80	2009/353	2009/341	1		Y
BERNDS.SEC14.HYD861	2009/338	082527	F	80	2009/353	2009/341	2		Y
BERNDS.SEC14.HYD861	2009/338	082638	F	80	2009/353	2009/341	3		Y
BERNDS.SEC14.HYD861	2009/338	082749	F	80	2009/353	2009/341	4		Y

STACKED VOLUME INFO

COPY EXPORT INFO

VOLSER	CURRENT LOCATION	DESTI NATION	IN TRAN	RETENTION DATE	V R	EXPORT DATE	EXPORT TIME
A02039	ATL3484F	MAZ1	Y	2020/001	Y	2009/338	083938
A02039	ATL3484F	MAZ1	Y	2020/001	Y	2009/338	083938
A02039	ATL3484F	MAZ1	Y	2020/001	Y	2009/338	083938
A02039	ATL3484F	MAZ1	Y	2020/001	Y	2009/338	083938

Overview

- EDGJCEXP Report – sorted **stacked volume**

Copy Exported Data Sets By Stacked Volume

- 1 -

12/08/2009

03:30:22

based on Bulk Volume Information Retrieval data

Stacked Volume Info: A02039 ATL3484F MAZ1 Y 2020/001 Y 2009/338 083938

LOGICAL VOLUME INFO			
VOLSER	VOLSEQ	REQUIRED LOCATION	EXPIRATION DATE
HYD861	1	MAZ2	2009/341
HYD861	1	MAZ2	2009/341
HYD861	1	MAZ2	2009/341
HYD861	1	MAZ2	2009/341

DATA SET INFORMATION									
DATA SET NAME		CREATE DATE	CREATE TIME	REC FM	BLK SIZE	RETENTION DATE	EXPIRATION DATE	PHYSICAL FILE	V SEQ R
BERNDS.MULTI.VOLUME.DS1		2009/338	082524	FB	80	2009/353	2009/341	1	Y
BERNDS.SEC14.HYD861		2009/338	082527	F	80	2009/353	2009/341	2	Y
BERNDS.SEC14.HYD861		2009/338	082638	F	80	2009/353	2009/341	3	Y
BERNDS.SEC14.HYD861		2009/338	082749	F	80	2009/353	2009/341	4	Y

Agenda

- z/OS Release 13 Enhancements
- Selected z/OS Release 12 Enhancements



- Appendix

DS8700



**TS1140
Tape Drive**



**TS7680
Deduplication
Gateway**



TS3500 Library



**TS7700
Virtualization
Engine**



Summary of z/OS DFSMSrmm V1R13

+: Support integrated into release base

<i>Function</i> / <i>z/OS (RMM) release</i>	<i>z/OS V1.13</i>	<i>z/OS V1.12</i>	<i>z/OS V1.11</i>	<i>z/OS V1.10</i>
<i>VRSELEXCLUDE & RM(EXPDT)</i>	+	OA32984 (Toleration)	OA32984 (Toleration)	OA32984 (Toleration)
<i>TS1140 Support</i>	+	OA33958	OA33958	OA35804
▪ Selective volume movement ▪ More „Last change“ details ▪ Last Reference Date for VRS ▪ ISPF Navigation Enhancements ▪ Show Effective Retention/Expiration Date ▪ Search Dataset Extensions ▪ TVEXTPURGE Extra Days ▪ More information on Expiry Date source ▪ Enhanced Tape Copy Support	+			
<i>Retention limit reporting</i>	+	+	OA30881	OA30881
<i>Volume Hold</i>	+	+	OA30436 (Honor Volume Hold)	OA30436 (Honor Volume Hold)
• EAS Eligibility • OPENRULE IGNORE • IPv6 • AUTOR • Addt. Status commands & RAS enhancements	+	+		
<i>Option to turn uppercasing on/off</i>	+	+	OA32661	OA32661
<i>TS7700 1.6 Support , Logical WORM</i>	+	+	OA28637	OA28637

ration

Important APAR and Enhancements Provided In the Service Stream

- **OA34036**
 - The “volume hold” flag –newly introduced with z/OS V1.12- will be honored on R10, R11
- **OA31661**
 - Additional option to control uppercasing of dataset names. The default is to convert to upper case (R9-R11)
- **OA32754, OA33498**
 - In a client/server configuration the client may hang after a network error (R9-R12)
 - Or a loop issuing EDG0356E SERVER COMMUNICATION ERROR messages may occur
- **OA33876 (currently open, R11)**
 - TS7700 COPY EXPORT related fixes and documentation updates
 - Fixes to address problems until microcode fix level is available
 - Refer to updated documentation:
<http://publibz.boulder.ibm.com/zoslib/pdf/OA33876.pdf>

Summary of z/OS DFSMSrmm V1R12

+: Support integrated into release base

<i>Function</i> \ <i>z/OS (RMM) release</i>	<i>z/OS V1.12</i>	<i>z/OS V1.11</i>	<i>z/OS V1.10</i>	<i>z/OS V1.9</i>
<i>Retention limit reporting</i>	+	OA30881	OA30881	
<i>Volume Hold</i>	+	OA30436 (Honor Volume Hold)	OA30436 (Honor Volume Hold)	
•EAS Eligibility •OPENRULE IGNORE •IPv6 •AUTOR •Addt. Status commands & RAS enhancements	+			
<i>Option to turn uppercasing on/off</i>	+	OA32661	OA32661	OA32661
<i>TS7700 1.6 Support , Logical WORM</i>	+	OA28637	OA28637	OA28637
•Report generator extensions •Journaling for D/R, EDGUPDT •EDGINERS SCAN	+	+		
<i>Migration checks for z/OS V1.11+ coexistence</i>	+	OA32028	OA26947 OA32028	OA26947 OA32028
<i>z/OS V1.11+ coexistence</i>	+	N/A	OA25714 OA28232	OA25714 OA28232

Where to go for more information on DFSMSrmm

- DFSMSrmm Homepage: <http://www.ibm.com/systems/storage/software/sms/rmm>
 - DFSMSrmm Enhancements: <https://www.ibm.com/support/docview.wss?q1=T1010391&rs=0&uid=isg3T1010391>
- z/OS V1.12 DFSMS library:
<http://www.ibm.com/systems/z/os/zos/bkserv/r12pdf/#dfsms>, especially
 - DFSMSrmm Managing and Using Removable Media
<http://publibz.boulder.ibm.com/epubs/pdf/dgt2r390.pdf>
 - DFSMSrmm Implementation and Customization Guide
<http://publibz.boulder.ibm.com/epubs/pdf/dgt2c890.pdf>
-  DFSMSrmm Primer: <http://www.redbooks.ibm.com/abstracts/SG245983.html>
- Contact the DFSMSrmm team: DFSMSrmm@de.ibm.com

Thank you!

धन्यवाद

Hindi

多謝

Traditional Chinese

ขอบคุณ

Thai

Спасибо

Russian

Gracias

Spanish

Thank You

English

Obrigado

Brazilian Portuguese

شكراً

Arabic

Grazie

Italian

多谢

Simplified Chinese

Danke

German

Bedankt

Dutch

Merci

French

நன்றி

Tamil

ありがとうございました

Japanese

감사합니다

Korean